

## Large Language Model Service

> Relevant source files

### Purpose and Scope

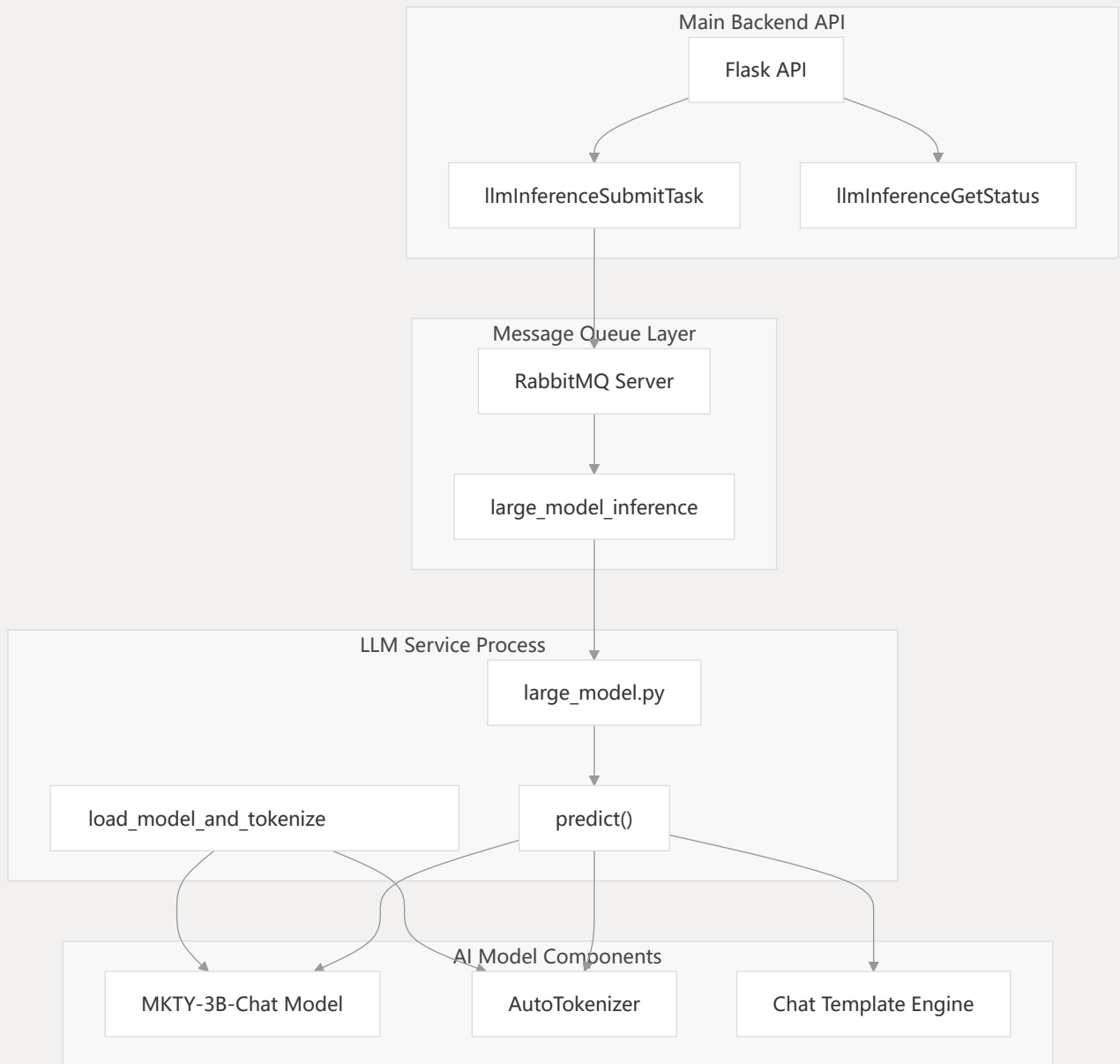
The Large Language Model Service provides medical Q&A capabilities and multi-agent discussion functionality using the MKTY-3B-Chat model. This service operates as a RabbitMQ message queue consumer that processes natural language inference requests asynchronously from the main backend API.

This document covers the LLM inference service implementation, message queue integration, and model loading mechanisms. For multimodal diagnosis capabilities combining text and images, see [Multimodal AI Services](#). For time series and text analysis features, see [Time Series & Text Analysis](#).

### Service Architecture

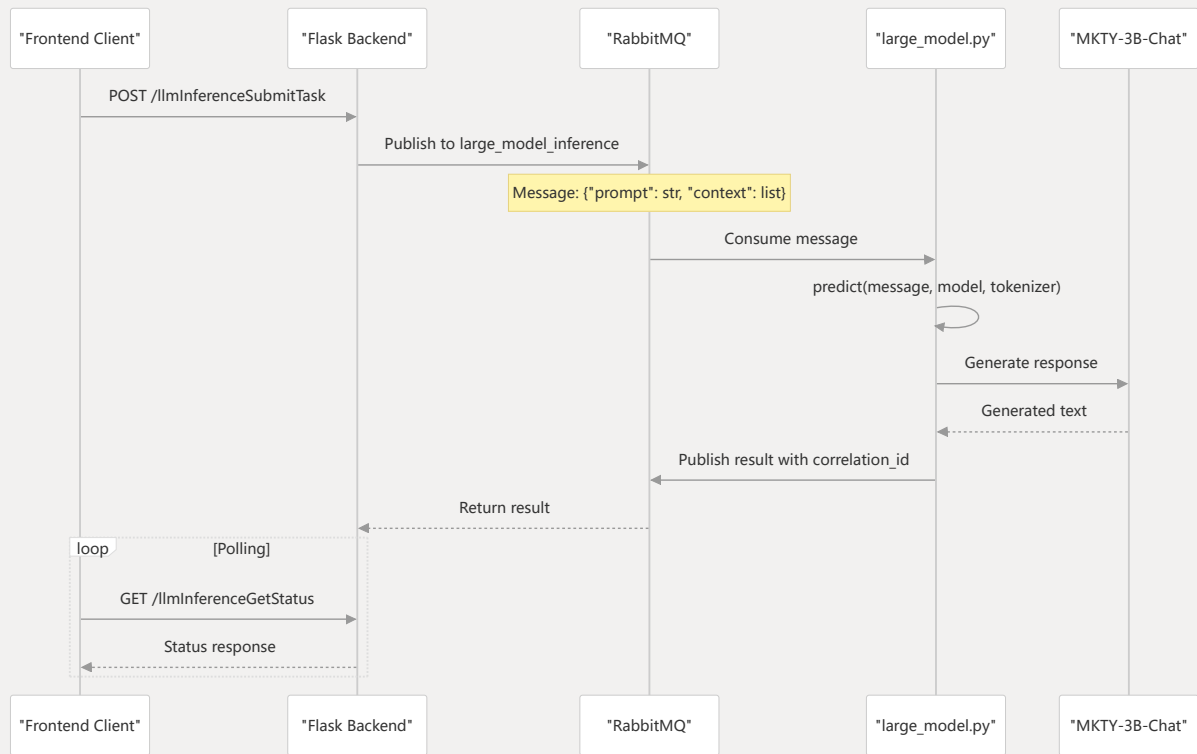
The Large Language Model Service follows a message queue-based architecture that decouples AI inference from the main web application, enabling scalable and asynchronous processing.

### System Integration Overview



Sources: `backend/large_model.py` 1-63 `backend/large_model_util.py` 66-95  
`backend/large_model_util.py` 97-116

## Message Queue Communication Flow



Sources: `backend/large_model.py` 26-58 `backend/large_model_util.py` 83-97

## Core Components

### Model Service Entry Point

The main service process is implemented in `large_model.py` and serves as the primary RabbitMQ consumer for LLM inference requests.

| Component                             | Function            | Configuration                             |
|---------------------------------------|---------------------|-------------------------------------------|
| <code>MQ_NAME</code>                  | Queue identifier    | <code>'large_model_inference'</code>      |
| <code>MODEL_PATH</code>               | Model directory     | <code>'./mkty'</code>                     |
| <code>MODE</code>                     | Runtime environment | <code>'dev'</code> or <code>'prod'</code> |
| <code>mq_connection_parameters</code> | RabbitMQ settings   | Host: localhost, Port: 5672               |

Sources: `backend/large_model.py` 10-19

### Inference Function Implementation

The `predict()` function handles the core model inference logic, processing conversational context and generating responses.



## predict() Function Flow

message: dict



Extract prompt & context



context.append(user\_mes



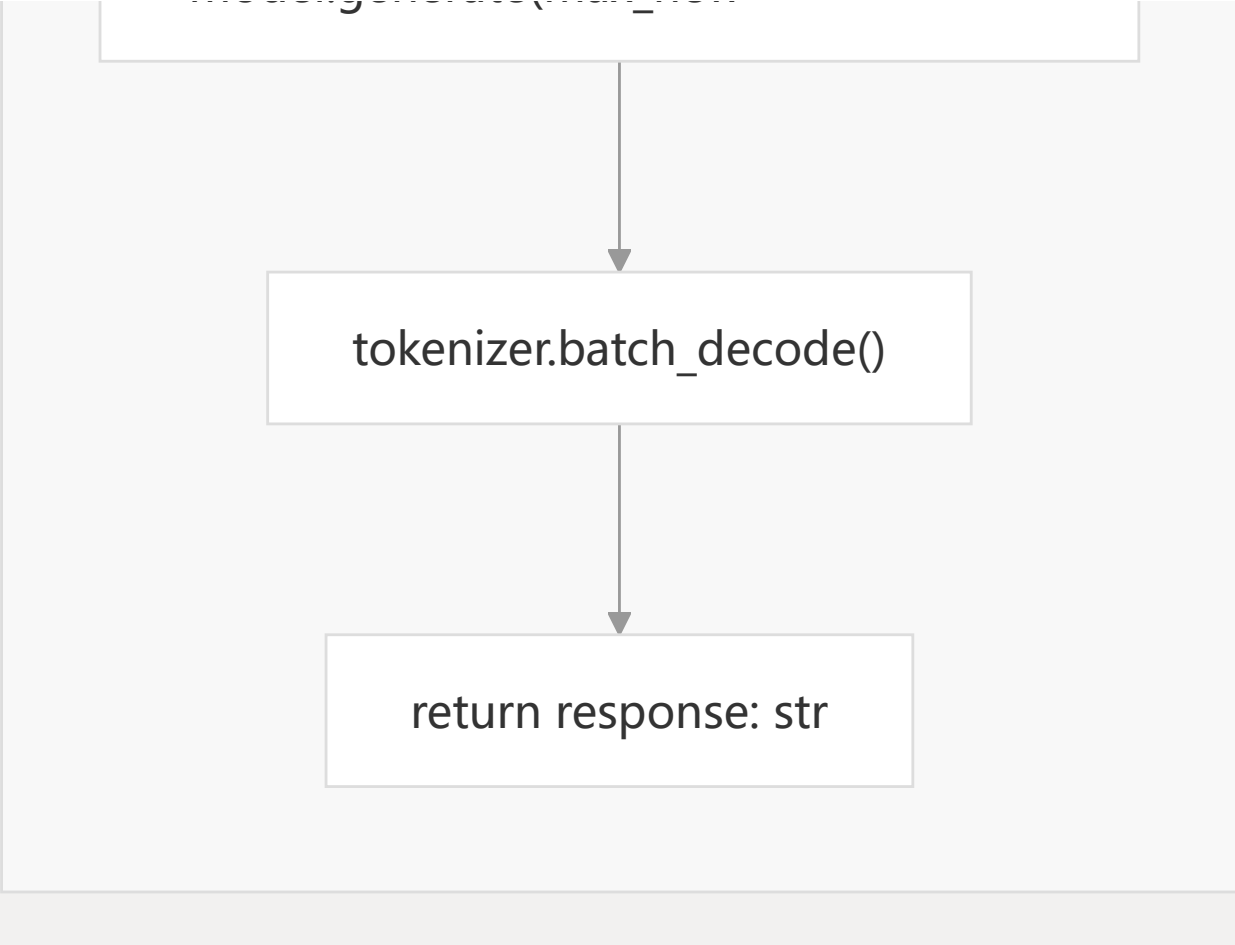
tokenizer.apply\_chat\_tem



tokenizer(text,  
return\_tensors='pt')



model.generate(max new



```
graph TD; A[ ] --> B[tokenizer.batch_decode()]; B --> C[return response: str];
```

tokenizer.batch\_decode()

return response: str

Sources: `backend/large_model.py` 26-58

## Model Loading and Initialization

The service uses Hugging Face transformers to load the MKTY-3B-Chat model with automatic device mapping and optimization.

| Parameter                   | Value               | Purpose                       |
|-----------------------------|---------------------|-------------------------------|
| <code>torch_dtype</code>    | <code>"auto"</code> | Automatic precision selection |
| <code>device_map</code>     | <code>"auto"</code> | Automatic GPU/CPU placement   |
| <code>max_new_tokens</code> | <code>2000</code>   | Maximum response length       |

Sources: `backend/large_model_util.py` 97-116

## Message Format and Protocol

### Input Message Structure

The service expects messages with the following JSON structure:

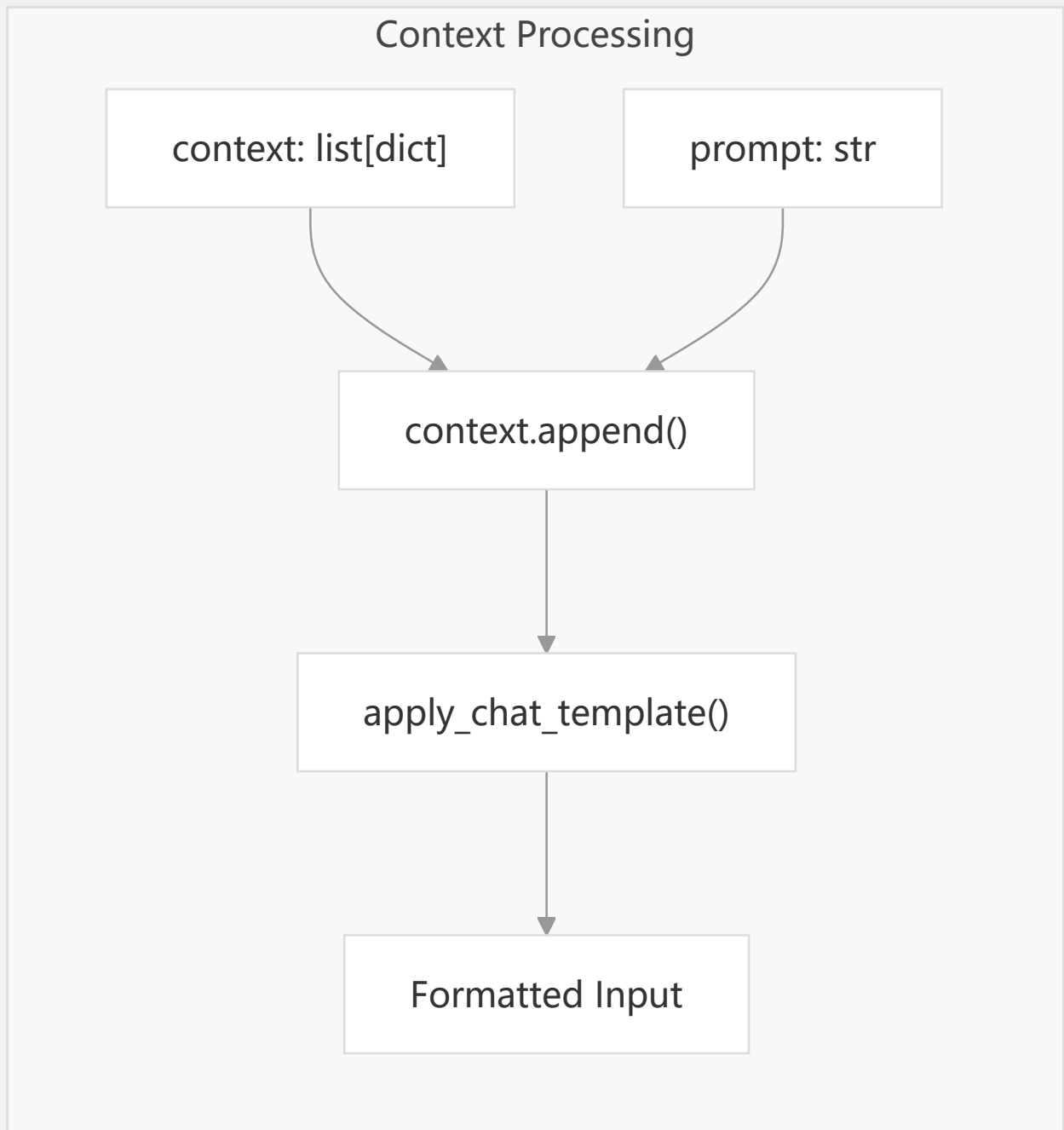
```
{
  "prompt": "string - Current user question",
  "context": [
    {"role": "user", "content": "Previous user message"},
    {"role": "assistant", "content": "Previous AI response"}
  ]
}
```

Sources: `backend/large_model.py` 31-41

## Conversation Context Management

The system maintains conversation history by appending the current prompt to the existing context array before processing.

## Context Processing



Sources: `backend/large_model.py` 39-46

## Utility Functions and Infrastructure

### Logging System

The service implements a rich console-based logging system with timestamp formatting and color-coded log levels.

| Log Level | Color | Usage               |
|-----------|-------|---------------------|
| info      | Green | General information |



Log Level

Color

Usage